



Obsidian

△ **Tumble-polished** obsidian piece

Obsidian is a natural volcanic glass that forms when lava solidifies so quickly that mineral crystals do not have time to grow. Technically, obsidian can have any chemical composition, although it is usually the product of silica-rich magmas. It is typically jet-black; hematite (iron oxide) contained within it can result in red and brown varieties, and the inclusion of tiny gas bubbles can create a golden sheen. In snowflake obsidian, clusters of light-coloured, needle-like crystals of cristobalite on broken surfaces resemble snowflakes.

Specification

Chemical name Silicon dioxide | **Formula** Composed of SiO_2 , MgO , and Fe_3O_4 | **Colours** Black, red, brown | **Structure** Amorphous | **Hardness** 5–6 | **SG** 2.35–2.60 | **RI** 1.45–1.55 | **Lustre** Vitreous | **Streak** White | **Locations** Europe, North America, South America, Australasia, Japan



Mexican obsidian | Rough | This gemmy, highly reflective piece of black obsidian originates from the central highlands of Mexico.



Snowflake obsidian | Rough | As this variety of obsidian cools, snowflake-like white crystals of cristobalite are formed, creating its distinctive appearance.



Rainbow obsidian | Cut | As with sheen obsidian (see right), rainbow obsidian contains small, oriented platelets, giving it an iridescence when polished.



Sheen obsidian | Cut | During its formation, platelets of other minerals form within the obsidian, giving it a sheen when polished, as seen here.



Cat carving | Carved | Although obsidian is brittle and glass-like, with careful carving, attractive ornaments can be produced from the material, such as this sculpture of a wistful cat in patterned snowflake obsidian. Its surface has been polished to a high shine.

Polished surface

Obsidian blades

Ancient cutting tools

When obsidian breaks, it can form an edge sharper than that of a steel scalpel. It was often used throughout antiquity to fabricate cutting tools and weapons, and was a prized material, widely traded across vast distances. It was used from the Stone Age onwards in civilizations including pre-Colombian Mesoamericans, ancient Egyptians, Native Americans, and others.

Obsidian spearhead and knife These blades originate from the Admiralty Islands off New Guinea from around 1900. They retain parts of their painted handles.

